

**Session 12: StructuresTopics:Declaring, initializing structures.Nested structures.Example:
Store student details (name, marks, grade).**

Tasks

1.

Declare a structure named Playlist to store details of a song: title (string), artist (string), and duration in seconds (integer). Initialize one Playlist variable with your favorite song's details and print each field.

```
#include <stdio.h>
```

```
struct Playlist
```

```
{  
    char title[100];  
    char artist[100];  
    int duration;  
};
```

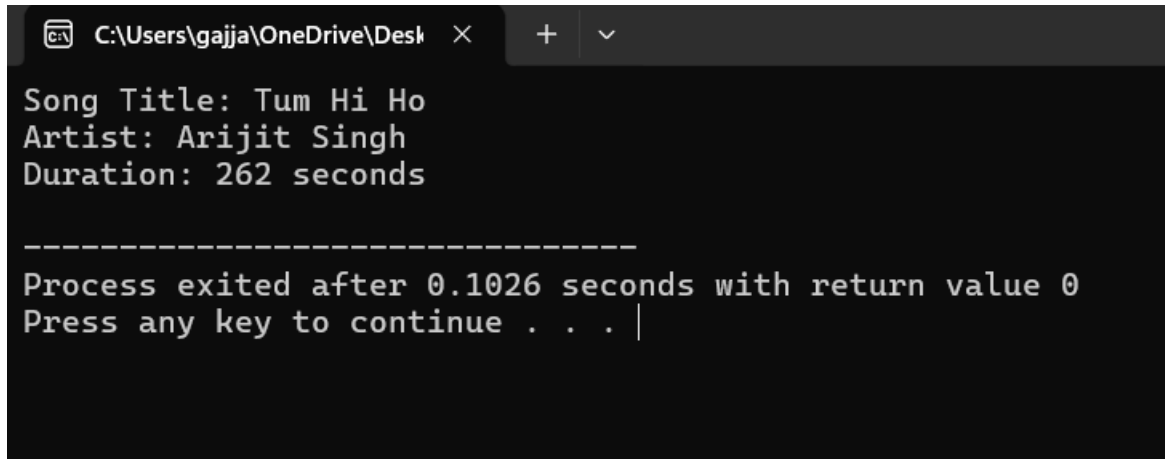
```
int main()
```

```
{  
    struct Playlist song = {  
        "Tum Hi Ho",  
        "Arijit Singh",  
        262  
    };
```

```
    printf("Song Title: %s\n", song.title);  
    printf("Artist: %s\n", song.artist);  
    printf("Duration: %d seconds\n", song.duration);
```

```
    return 0;
```

```
}
```



```
C:\Users\gajja\OneDrive\Desktop × + v
Song Title: Tum Hi Ho
Artist: Arijit Singh
Duration: 262 seconds

-----
Process exited after 0.1026 seconds with return value 0
Press any key to continue . . . |
```

2.

Create a structure called `FoodItem` to store Zomato-style menu data: `itemName` (string), `price` (float), and `rating` (float). Initialize an array of 3 `FoodItem` variables with real menu items and display their details using a loop.

```
#include <stdio.h>
```

```
struct FoodItem
```

```
{
```

```
    char itemName[100];
```

```
    float price;
```

```
    float rating;
```

```
};
```

```
int main()
```

```
{
```

```
    struct FoodItem menu[3] =
```

```
    {
```

```
        {"Paneer Pizza", 249.00, 4.5},
```

```
        {"Veg Burger", 149.00, 4.2},
```

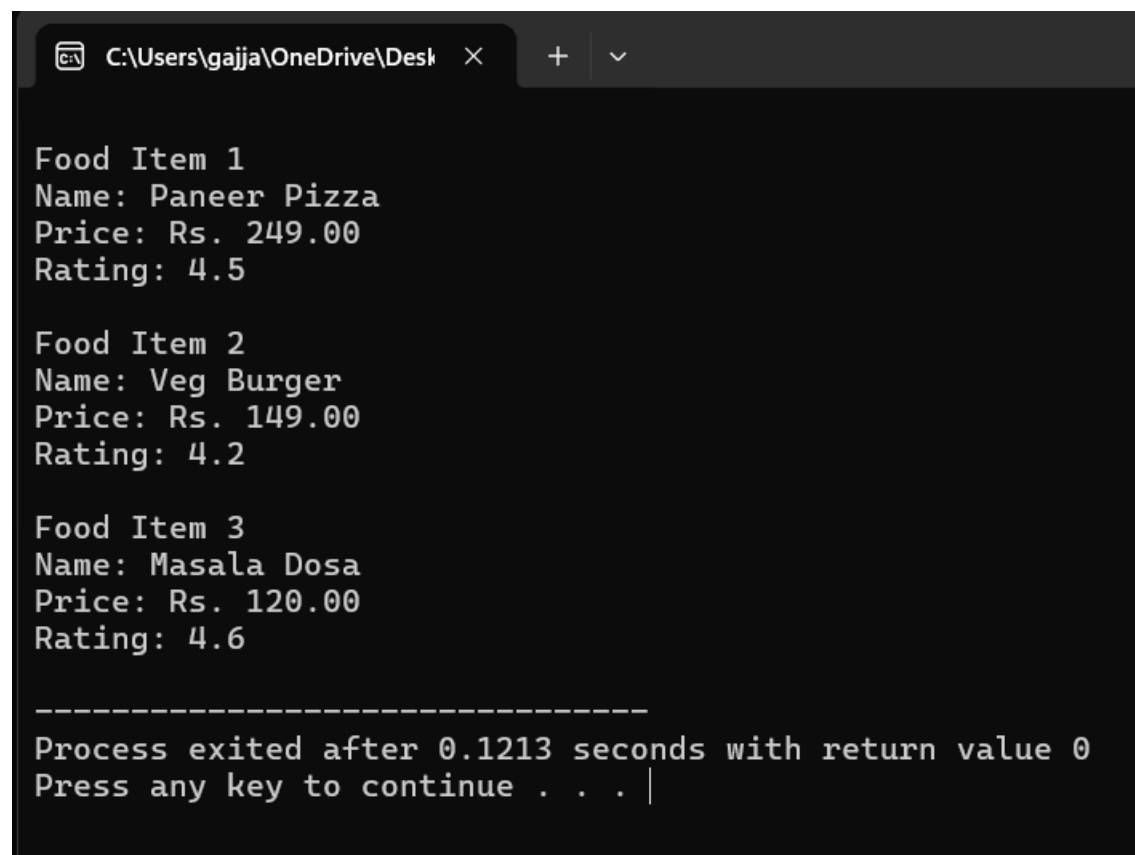
```
        {"Masala Dosa", 120.00, 4.6}
```

```
};

int i;

for(i = 0; i < 3; i++)
{
    printf("\nFood Item %d\n", i + 1);
    printf("Name: %s\n", menu[i].itemName);
    printf("Price: Rs. %.2f\n", menu[i].price);
    printf("Rating: %.1f\n", menu[i].rating);
}

return 0;
}
```



```
C:\Users\gajja\OneDrive\Desktop >
Food Item 1
Name: Paneer Pizza
Price: Rs. 249.00
Rating: 4.5

Food Item 2
Name: Veg Burger
Price: Rs. 149.00
Rating: 4.2

Food Item 3
Name: Masala Dosa
Price: Rs. 120.00
Rating: 4.6

-----
Process exited after 0.1213 seconds with return value 0
Press any key to continue . . . |
```

3.

Define a nested structure called MovieShow for a BookMyShow-style app: Movie (string), Screen (integer), and a nested structure Time with hours and minutes (integers). Create and initialize a MovieShow variable for any movie and print its details in the format 'Movie: X, Screen: Y, Time: HH:MM'.

```
#include <stdio.h>
```

```
struct Time
```

```
{
```

```
    int hours;
```

```
    int minutes;
```

```
};
```

```
struct MovieShow
```

```
{
```

```
    char Movie[100];
```

```
    int Screen;
```

```
    struct Time time;
```

```
};
```

```
int main()
```

```
{
```

```
    struct MovieShow show =
```

```
    {
```

```
        "Avengers",
```

```
        3,
```

```
        {18, 30}
```

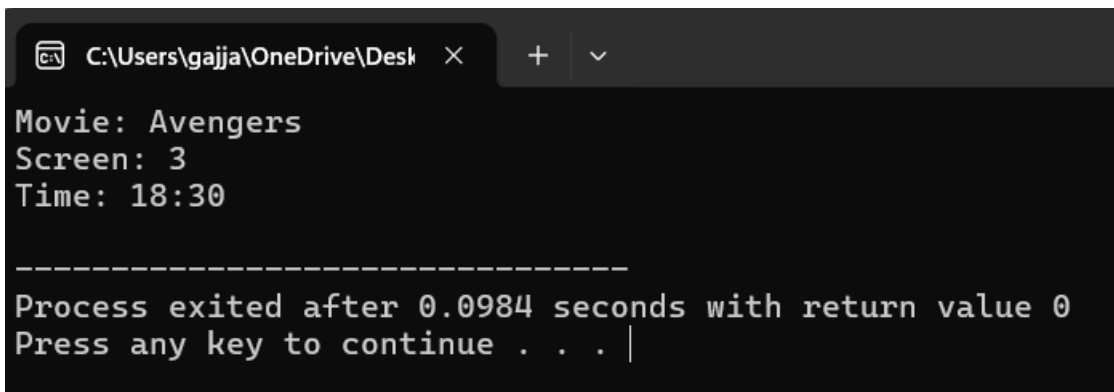
```
    };
```

```

printf("Movie: %s\n", show.Movie);
printf("Screen: %d\n", show.Screen);
printf("Time: %02d:%02d\n",
       show.time.hours,
       show.time.minutes);

return 0;
}

```



```

C:\Users\gajja\OneDrive\Desktop >
Movie: Avengers
Screen: 3
Time: 18:30
-----
Process exited after 0.0984 seconds with return value 0
Press any key to continue . . . |

```

4.

Build a structure called InstaProfile with fields: username (string), followers (integer), and a nested structure Bio with fields: description (string) and age (integer). Initialize an InstaProfile variable with your own details and display all fields.

```
#include <stdio.h>
```

```
struct Bio
```

```

{
    char description[200];
    int age;
};

```

```
struct InstaProfile
```

```
{
```

```
char username[100];

int followers;

struct Bio bio;

};


int main()

{
    struct InstaProfile profile =
    {
        "manan_gajjar",
        5000,
        {
            "Software Developer and MCA Student",
            22
        }
    };


    printf("Username: %s\n", profile.username);
    printf("Followers: %d\n", profile.followers);
    printf("Description: %s\n", profile.bio.description);
    printf("Age: %d\n", profile.bio.age);


    return 0;

}
```

C:\Users\gajja\OneDrive\Desk × + ▾

Username: manan_gajjar

Followers: 5000

Description: Software Developer and MCA Student

Age: 22

Process exited after 0.09365 seconds with return value 0

Press any key to continue . . . |